SIMATS ENGINEERING

ASSESSMENT TOOL 2 : Scenario-Based

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments.* (BL3)

Dave's Taxonomy: Manipulation (Level 2)
Question Count: 5

Total Marks: 25

Code of Conduct

I K·Mahesh babu (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.
Signature of the student: KMaheshbabu

Assessment Tasks
Scenario-Based Questions

1. A media company wants to migrate its video streaming platform to the cloud.
   The system must handle millions of concurrent users with minimal buffering.
   Design a cloud architecture using scalability and caching techniques.
   Explain how load balancing and auto-scaling can ensure smooth streaming.
   Discuss cost optimization strategies for such high-demand workloads.

2. A fintech startup is deploying applications using containers across multiple cloud providers.
   They face challenges in maintaining consistency and managing deployments.
   Propose a solution using container orchestration and multi-cloud strategies.
   Explain how service discovery and scaling can be handled efficiently.
   Highlight the benefits and risks of a multi-cloud deployment model.

3. An e-learning platform experiences data loss due to accidental deletion in the cloud.
   The organization needs a robust backup and disaster recovery strategy.
   Design a solution using cloud-native backup, replication, and versioning.

Explain how recovery time objective (RTO) and recovery point objective (RPO) are achieved.
Provide steps to ensure business continuity during failures.

4. A government agency is adopting a hybrid cloud model for sensitive and non-sensitive data. Some applications must remain on-premises due to compliance requirements.
   Design a hybrid architecture that ensures secure communication between environments.
   Explain how identity management and encryption should be implemented.
   Discuss challenges in managing hybrid cloud environments.

5. A SaaS provider needs to ensure high availability and fault tolerance for its application. The system must continue operating even if one region goes down.
   Propose a multi-region deployment strategy in the cloud.
   Explain how data replication and failover mechanisms should be configured.
   Discuss monitoring and alerting strategies for proactive issue detection.

Scenario-Based Rubric

Criteria	**Weightage**	**Level 4 - Exemplary**	**Level 3 - Proficient**	**Level 2 - Developing**	**Level 1 - Beginning**
Understanding of Scenario	20%	Demonstrates complete understanding of the scenario context	Good understanding with minor gaps	Partial understanding	Fails to understand the scenario
Identification of Risks / Issues	20%	Identifies all major issues correctly	Identifies most issues	Limited issue identification	Misses major issues
Application of Concepts	25%	Applies virtualization concepts effectively	Applies concepts with minor mistakes	Partially correct application	Weak application
Decision Justification	20%	Decisions are well justified and technically strong	Reasonable justification	Weak justification	No useful justification
Clarity	15%	Clear and well-organized response	Generally clear	Somewhat unclear	Poorly presented

Cloud Auto-Scaling and Load balancing:

Cloud Computing Enable bussiness to handle Every. Workloads Efficiently. During seasonal sale such as black-friday, Diwali, (or) festival offers, e-commerce websites Experience Sudden traffic spikes. if the application, cannot handle the increased number of users, it may become slow (or) cash. To over Come. problem, Cloud Providers offer auto-Scaling and load balancing.

Auto Scaling:

Automatically increases (or) decreases the number of Virtual Severs based on user demand. when website traffic increases, new Servers instances are Created automatically. When traffic decreases, the Extra Servers are removed This helps maintain application performance while reducing unecessary costs.

Load balancing:

Distributes incoming user requests evenly among multiple Servers. Instead of Sending all requests to one Server, the load balancer forward Each request to the least busy (or) avaliable Server. This Prevents overload and Ensures that all servers are used Efficiently.

The Combination of auto-scaling and load balancing Significantly reduces downtime. If one Server fails, the load balancer redirects traffic to heavily Servers, Ensuring uninterrupted service. Customer Experience faster page loading, quick checkout, and Smooth online shopping.

Monitoring tools:-

Such as Cloud watch, Azure monitor, & Google Cloud Monitoring Continuously monitor cpu usage, memory utilization, response time, and network traffic. when resources usage Exceeds Predefined limits, they automatically trigger auto-Scaling and notify administrators about any issues.

Conclusion:-

Auto-scaling, load balancing, and monitoring work togther to improve applications Performance, reduce down-time, optimize resource usage, and provide a smooth user Experience during high-traffic Events.

Scenario:

An organization wants to store large Volumes of backup data securely in the cloud. The organization requires a storage Solution that Provides high durability strong Security, low cost, and quick recovery in case of data loss (or) disasters.

1. Cloud Storage Ensuring Durability and data Redundancy:

Cloud storage Ensures data durability by storing multiple Copies of backup data on different servers and in multiple geographical locations. This process is called data redundancy. If one server, storage device, or data Center fails due to hardware failure, or natural disasters, another copy of the data is immediatly avaliable.

2. Protection using Encryption and Access control:

To Protect sensitive backup data, cloud Providers use Encryption and access control. Encryption converts data into an unreadable format while it stored (Encryption) and while it transferred over the network. only users with the Correct Encryption key can access the original data. Access control is implemented using identity and Access Management (IAM), role-based Permissions, and Multi-factor Authentication (MFA). These security measure ensure that only authorized Employees can access (or) store backup, Preventing unauthorized data

5. **Cost Advantages Compared to Traditional storage:-**

Cloud storage is more cost-Effective than traditional storage Systems because organizations do not need to Purchase Expensive Storage hardware (or) maintain their own data centers. Expences related to Electricity Cooling, maintenance, and hardware upgrades are Eliminated. cloud Providers follow as pay-as you-use Pricing model, allowing organizations to pay for the storage they actually use.

3. **Content Delivery Networks (CDNs) and Global Deployment:**

Scenario:-

- A Mobile app Company plans to deploy its application globally.
- The Company wants to users from different Countries to access the app with high speed, low latency, and uninterrupted Service.
- It uses Content delivery Networks (CDNs) and cloud Services to achieve this.

How CDNs improve Performance and reduce latency:-

- A CDN is a network of Servers located in different regions.
- It stores copies of frequently used content such as image s, Videos and, applications files
- It reduces the distance data travels

How Edge Servers Enhance user Experience:-

- Edge Servers are located close to Servers.
- They Stored Cached copies of popular Content.
- They Provide faster loading and quicc response time.
- They reduce buffering and network delays.
- They improve the user Experience Even during peak traffic.

How cloud Providers Support Global Avaliability:-

- cloud Providers like Aws, Microsoft Azure and Google Cloud have data Centers world wide.
- They use load balancing to distribute user requests.
- They use data replication to Keep copies of data in multiple Goals
- This Ensures 24/7 availability and reliable Savie

Conclusion:-

- CDNs improve application Speed and reduce latency
- Edge Servers deliver Content quickly to users.
- Cloud Providers ensure global avaliability and realiability.
- Together, they Provide a fast, smooth, and the uninterrupted Experience for users across the world

4. Cloud Security, compliance and Auditing:-

Scenario:-

- A financial insitution stores customer and transaction data in the cloud.
- It requires strict security, regulatory compliance and Protection of Sensitive financial information

How cloud Providers Support Regulatory Compliance & Auditing:-

- Cloud providers Support standards Such as PCI DSS, ISO 27001, GDPR, and SOC 2.
- They Provide Security Controls, compliance reports, and regular Security audits.
- Audits trails help track all user activity and System changes

Role of logging, Monitoring and Encryption:-

- logging records users logins, transactions and System activities.
- Monitoring continuously checks the System for Suspicious activities and Security threats.
- Encryption Protects data both at rest and in in-transmit.
- They feature Prevent unauthorized access and Ensure Customer data remains Secure.

Ensuring data Sovereignty:-

- The company can choose a cloud region where customer data is stored.
- Data remains within the required country to meet local government regulations.
- Regular audits Ensure compliance with the legal requirements.

5. Containers, Kubernetes and Micro Service:-

Scenario:-

- A development team want to use Containeri-zation for deploying applications.
- The goals is to improve Performance, simplify deployment, and run the applications across different cloud Platforms.

1. Difference between Containers and Virtual machine (VMs):-

- Containers share the host ~~operating~~ operating System and Package only the applications and its the dependencies.
- Virtual Machines [VMs] have a separate operating system for Each instance.
- Containers are light weight, start faster, and use less cpu, memory and Storage.

How Kubernetes Simplifies Deployment and Scaling:-

- Kubernetes is a Container Orchestration Platform.
- It automates the deployment and management of containers.
- It automatically scales applications based on user traffic.

Probability Access across different cloud Environments:-

- Containers package the applications with all required libraries and dependencies.
- The Same Container can run on Aws, Microsoft Azure, Google Cloud, & private cloud without changing the code.
- This Provides easy migration, flexibility, and avoids Vendors lockin.

Conclusion:-

- Containers Provide better Performance and use fewer resource than Virtual machines.
- Kuberrettes Simplifies deployment, scaling, and managment of containerized applications.
- Container Portability allows applications to run across multiple cloud Platforms.